Massen Esmaeili

Ph.D. Candidate, Computational Modeling & Simulation – University of Pittsburgh

moe32@pitt.edu | Pittsburgh, PA, US | LinkedIn | Google Scholar

Research focus: Quantum-inspired tensor-network methods (matrix product states, MERA, PEPS) for reduced-order modeling and high performance computing of turbulent and reacting flows, and near-term quantum algorithms for fluid-dynamics PDEs, bridging quantum computing and computational fluid dynamics.

Education

Ph.D., Computational Modeling & Simulation (Mechanical Engineering) 2023 – Present
University of Pittsburgh, Pittsburgh, PA | *Advisor: Prof. Peyman Givi*

- GPA: 4.00/4.00.
- Research: quantum-inspired tensor-network encodings for reduced-order modeling of turbulent flows
- Relevant coursework: Tensor Networks (A); Quantum Computing (A); Reduced-Order Modeling for Engineers (A+); Turbulence (A); Numerical Methods in Scientific Computing (A+).

M.S., Mechanical Engineering – Computational & Data-Enabled Engineering (CDEE) Track 2023 – 2026
John A. Swanson School of Engineering, University of Pittsburgh, Pittsburgh, PA

- GPA: 4.00/4.00.

B.Sc., Mechanical Engineering 2017 – 2022
Sharif University of Technology, Tehran, Iran

- CGPA: 3.73/4.0 | Final two years GPA: 4.0/4.0

Publications & Presentations

Journal articles & preprints

- **M. Esmaeili**, H. Alipanah, R. Pinkston, P. Givi, D. Livescu, A. J. Daley, D. Jaksch, J. J. Mendoza Arenas. *A Priori Assessment of Tensor-Network Encoding for Reduced-Order Modeling of Isotropic Turbulent Flows.* arXiv:2608.28869, 2026. (under review, *Journal of Fluid Mechanics*)

Conference papers & presentations

- **M. Esmaeili**, K. Wang, *et al.* **Tensor Networks for Reduced-Order Modeling of a Turbulent Jet Flame.** Oral presentation (presenting author), 79th Annual Meeting of the APS Division of Fluid Dynamics (DFD), Orlando, FL, November 2026. (Upcoming) [abstract]
- Daniel Madrid, **M. Esmaeili**, *et al.* **Cross-Platform Comparison of Quantum Hardware for Simulation of the Advection–Diffusion Equation.** 79th Annual Meeting of the APS Division of Fluid Dynamics (DFD), Orlando, FL, November 2026. (Upcoming) [abstract]
- **M. Esmaeili**, K. Wang, H. Alipanah, R. Pinkston, D. Madrid, *et al.* **Low-Rank Tensor Network Ansätze for Reduced-Order Modeling of a Non-Premixed Turbulent Flame.** Annual Meeting of the New York–New Jersey–Pennsylvania Section of SIAM (SIAM-NNP), October 2026. [link]
- R. Pinkston, **M. Esmaeili**, H. Alipanah, N. Gourianov, D. Jaksch, D. Livescu, P. Givi, J. J. Mendoza Arenas. **Matrix Product State for Computational Turbulent Scalar Mixing and Reaction.** Oral presentation (Minisymposium MS262: *Advances of Quantum and Quantum-Inspired Algorithms for Solving Flow Simulation Problems*), WCCM–ECCOMAS 2026, Munich, Germany, July 2026. [abstract]
- **M. Esmaeili**, H. Alipanah, R. Pinkston, P. Givi, D. Livescu, D. Jaksch, J. J. Mendoza Arenas. **A Priori Assessment of Tensor Networks Encoding for Reduced-Order Modeling of Isotropic Turbulence.** Oral presentation, 78th Annual Meeting of the APS Division of Fluid Dynamics (DFD), Houston, TX, November 2025. [abstract]

Research Experience

Graduate Research Assistant – University of Pittsburgh 2022 – Present
Advisor: Prof. Peyman Givi | *Collaborator: Prof. Juan José Mendoza-Arenas*

- Developed tensor-network (MPS/MERA/PEPS) pipelines to compress and reconstruct 1024^3 forced isotropic turbulence DNS fields (Johns Hopkins Turbulence Database), achieving controlled accuracy–compression trade-offs through singular-value truncation; first-author manuscript under review at *Journal of Fluid Mechanics*.

- Designed and benchmarked interleaved, split, and comb tensor orderings to minimize inter-tensor correlations, then validated reconstructions against DNS via low- and high-order turbulence statistics (gradient PDFs, strain-rate invariants, Q–R joint statistics).
- Extended the framework to turbulent non-premixed jet-flame data (CO/H_2 chemistry, $864 \times 1008 \times 576$ grid) for scalar mixing and reaction, developing low-rank MPS and PEPS ansätze and comparing against LES–FDF (AMReX/PeleLM) outputs.
- Contributed to a cross-platform study of quantum hardware for simulating advection–diffusion.
- Built parameterized SLURM workflows on the Pitt Center for Research Computing cluster (CRC), sweeping bond dimension across parallel analysis jobs (compression ratios, scatter/PDF/contour diagnostics).
- Implemented Julia (ITensors) and Python (quimb, TensorLy) routines for truncation, fidelity evaluation, and field-level reconstruction.
- Collaborating with Los Alamos National Laboratory and international quantum-physics groups; 5 conference presentations (2025–2026) and 1 preprint to date.

B.Sc. Thesis Researcher – Sharif University of Technology 2021 – 2022
Advisor: Prof. Navid Arjmand

- Modeled the biomechanical effects of lumbar fusion surgery on adjacent spinal segments using musculoskeletal models (AnyBody).

Technical Skills

Tensor networks	ITensors.jl, quimb, TensorLy, TensorCrossInterpolation; MPS, MERA, PEPS, TCI
Quantum computing	Qiskit; MPS-based quantum state preparation, variational algorithms (VQE)
Programming	Python (NumPy/SciPy), Julia, MATLAB, C/C++
HPC	SLURM workflow design, parallel batch pipelines, Pitt CRC, conda environments
Numerical methods	SVD / low-rank approximation, finite-element & finite-difference schemes, spectral methods, entropy-stable SBP–SAT solvers
Scientific computing	DNS/LES post-processing, turbulence statistics, data visualization

Professional Service & Affiliations

- Peer reviewer, *Physical Review Applied* (American Physical Society), 2026.
- Member, U.S. Association for Computational Mechanics (USACM) Student Chapter, University of Pittsburgh.
- Board Member, Engineering Graduate Student Organization (EGSO) and International Engineering Students' Organization (IESO), University of Pittsburgh.

Certifications & Professional Development

- **Womanium & WISER 2025 Quantum Program — Quantum Solvers: Algorithms for the World's Hardest Problems**, The Washington Institute for STEM, Entrepreneurship and Research, August 2025. Intensive training in quantum algorithms for differential equations and CFD. [verify]

Teaching

- Teaching Assistant, University of Pittsburgh (2023–2024) — ME2232 Mathematics of Data-Enabled Science & Engineering; ME0031 Electrical Circuits.
- Teaching Assistant, Sharif University of Technology (2017–2022) — Fluid Mechanics II, Heat Transfer I, Dynamics of Machinery, Die Design, FEA with ABAQUS, CAD with SOLIDWORKS; graded 1000+ student submissions and developed course material.

Honors & Awards

- Ranked 49th nationally (Math & Physics university entrance exam), among 150,000+ participants, 2017.
- Ranked 19th nationally (English entrance exam) and 10th nationally (Fine Arts entrance exam), 2017.
- Certified, National Mathematical Olympiad (2013, 2015); National Olympiad in Informatics (2015).
- Graduate of NODET (National Organization for Development of Exceptional Talents), Iran.